

Nonlinear feedback selects an action only while its energy source supplies power

An oscillator with a finite energy depot relaxes to zero action in the model below. Replenishing the depot above a threshold produces a positive attracting action, independent of the initial seed but dependent on the supplied power and oscillator frequency. The exact construction separates attraction from universality and gives a concrete test of a self-excitation mechanism.

Status: exploratory written derivation, Q03; independent proof/literature review is required before accepted-ledger promotion. This is an autonomous dissipative effective model, not a derivation from a closed Hamiltonian field.

1. Construction and physical choice

Let an oscillator have angular frequency $\omega > 0$, action $J \geq 0$ and angle θ . Its measured mechanical energy is $H = \omega J$. Let $e \geq 0$ be stored depot energy. Choose loss rates $\gamma, \kappa > 0$, conversion coefficient $g > 0$ with units (energy time) $^{-1}$, and supplied power $P \geq 0$. Define

$$\dot{\theta} = \omega, \quad \dot{J} = 2(ge - \gamma)J, \quad \dot{e} = P - \kappa e - 2ge\omega J. \quad (1)$$

The nonlinear transfer $2ge\omega J$ enters oscillator energy and leaves the depot with equal magnitude. The premise is stimulated gain proportional to existing oscillation energy: a charged or active reservoir amplifies a seed, while conversion depletes its stored energy. This is a deliberately simple feedback law to test, rather than a universal mechanical law.

A real-quadrature realization removes the angle singularity at rest. With coordinates Q, R of units square root of action and $J = (Q^2 + R^2)/2$, put

$$\dot{Q} = (ge - \gamma)Q - \omega R, \quad \dot{R} = \omega Q + (ge - \gamma)R.$$

These equations induce (1), including the invariant state $Q = R = 0$. For a harmonic mechanical readout take $x = Q/\sqrt{m\omega}$ and $p = \sqrt{m\omega}R$; then $p^2/(2m) + m\omega^2 x^2/2 = \omega J$. Gain acts on both quadratures, so this realization is an effective amplitude model rather than Newton's unmodified relation $\dot{x} = p/m$.

The physical idea has precedent: Schweitzer, Ebeling and Tilch describe particles that store environmental energy and convert it into motion, with limit cycles above an uptake threshold. Their model motivates the comparison; equation (1) is the explicit model tested here.

2. Finite fuel: an exact energy inequality

The positive quadrant is invariant. For $E = \omega J + e$, cancellation of the internal transfer gives

$$\dot{E} = P - 2\gamma\omega J - \kappa e. \quad (2)$$

Put $a = \min(2\gamma, \kappa) > 0$. At $P = 0$,

$$0 \leq E(t) \leq E(0)e^{-at}, \quad 0 \leq J(t) \leq E(0)e^{-at}/\omega. \quad (3)$$

Thus even an initially amplifying depot, with $ge(0) > \gamma$, eventually exhausts its energy and sends $J(t)$ to zero. For arbitrary fixed P , the bound $\dot{E} \leq P - aE$ also gives global bounded solutions. The conclusion depends explicitly on positive losses. Lossless closed dynamics is a different candidate, whose conserved energy can retain preparation dependence.

3. Replenishment: an explicit positive attractor

The quiet equilibrium is $(J, e) = (0, P/\kappa)$. The linear action growth rate there is $2(gP/\kappa - \gamma)$. Define

$$P_c = \frac{\kappa\gamma}{g}.$$

For $P > P_c$ a positive equilibrium exists:

$$e_* = \frac{\gamma}{g}, \quad J_* = \frac{P - P_c}{2\gamma\omega}. \quad (4)$$

It attracts every initial state with $J(0) > 0$ and $e(0) > 0$. To prove this, use the nonnegative function

$$V = \omega [J - J_* - J_* \log(J/J_*)] + e - e_* - e_* \log(e/e_*).$$

Writing $\delta J = J - J_*$ and $\delta e = e - e_*$, the first term contributes $2g\omega\delta J\delta e$. The depot equation is

$$\dot{e} = -(\kappa + 2g\omega J_*)\delta e - 2g\omega e\delta J.$$

Multiplication by $1 - e_*/e$ cancels the cross term exactly, giving

$$\dot{V} = -(\kappa + 2g\omega J_*) \frac{(e - e_*)^2}{e} \leq 0. \quad (5)$$

Sublevel sets of V are compact within the positive quadrant. On $\dot{V} = 0$ we have $e = e_*$; remaining there requires $J = J_*$. The invariant-set argument therefore proves convergence to (4). An initially empty depot becomes positive immediately when $P > 0$. An exactly zero oscillator seed stays zero even above threshold; the attracting positive branch has the basin $J(0) > 0$.

The result supplies an actual physical-time action attractor in a classical effective model. It also identifies precisely what powers it.

4. Selection tests and consequence

At the same g, γ, κ, P , probes of different frequency acquire the same energy $\omega J_* = (P - P_c)/(2\gamma)$, rather than the same action. Demanding a common action $K > 0$ instead requires the power prescription

$$P(\omega) = P_c + 2\gamma\omega K. \quad (6)$$

This prescription inserts the desired scale in the source. It is the physical step a further theory must explain, rather than a consequence of attraction. Moreover $J_* \downarrow 0$ continuously as $P \downarrow P_c$: instability of the quiet state above threshold supplies no positive lower action bound across the admitted source parameters. Exact rest remains a trajectory throughout.

The promising ingredient is feedback that erases initial amplitude; the failed ingredient is identifying its source-dependent attractor with a universal constant. Together with Q02's proposed frequency-selective radiation response, this suggests a sharper question: can a self-consistent reservoir dynamically set a common spectral amplitude, with its energy budget included? Merely connecting this depot to a linear radiation bath would still supply P .

Park further pumped-oscillator variants. A different route worth testing is a conservative field sector with a fixed topological constraint: its quiet-state exclusion would be a geometric restriction rather than continuous fueling. The decisive test is whether its minimum action survives dilation, not merely whether a nontrivial stationary configuration exists. This is a proposed next test, not evidence that topology already fixes an action constant.

Source and review boundary

F. Schweitzer, W. Ebeling and B. Tilch, *Complex Motion of Brownian Particles with Energy Depots*, Physical Review Letters **80**, 5044 (1998), publisher record and abstract. Coverage: one discovery query and the publisher abstract; no full-text equation match or novelty assessment. Its energy-uptake/limit-cycle mechanism prompted the finite-fuel versus replenished comparison. Equations (1)–(6) are the present exploratory calculation, with energy accounting and a written Lyapunov check; no numerical or symbolic scripts were used.